

Read quality and genome alignment in the PDAC KRAS dataset

WEEK 2 PROGRESS REPORT

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Methods revision: full alignments, quality checks and strand comparison.

Wording revised on 23 September 2026. Results retain the original reporting stage.

Purpose and progress

This stage assesses read quality and genome alignment before gene counting. All 17 full-run HISAT2 summaries are available. Together, the runs contain 727,731,715 read pairs.

Methods

FastQC 0.12.1 assessed the 34 input files. MultiQC 1.35 combined 32 Galaxy reports with two reports from the complete local SRR24828471 files. HISAT2 2.2.3 aligned the paired reads to human hg38 with Phred+33 quality scores, unstranded mode, no read limit and no end trimming. Other alignment options used the defaults.

The paired files were checked by sample and run accession. Incomplete imports were replaced with recovered files. Base-quality and adapter checks passed, so Cutadapt was not used.

Strand setting

RSeQC Infer Experiment sampled 200,000 reads with a minimum mapping quality of 30 using an hg38 BED12 annotation. The two orientation fractions were 46.90% and 46.73% for Control M1, and 45.80% and 45.69% for KRAS P1. These similar fractions support unstranded alignment for the two inputs. The remaining runs used this setting without individual strand checks.

Full alignment results

Sample	Input pairs	Overall (%)	Unique pairs (%)	Discordant (%)
Control M1	40,237,116	96.75	86.11	0.61
Control M2	45,764,716	96.78	85.19	0.58
Control M3	38,181,985	96.67	87.14	0.64
Control P1	45,946,458	96.73	87.25	0.57
Control P2	39,989,452	95.79	86.47	1.15
Control P3	47,840,900	96.41	87.03	0.54
Control P4	46,287,075	96.20	87.22	0.65
Control P5	45,023,981	96.75	88.32	0.57
KRAS M1	45,292,678	96.51	0.0030	75.56
KRAS M2	44,032,049	96.74	85.25	0.62
KRAS M3	42,961,283	96.64	86.95	0.60
KRAS P1	39,963,282	96.52	87.11	0.62
KRAS P2	42,632,214	96.49	87.50	0.69
KRAS P3	42,073,244	96.57	86.11	0.56
KRAS P4	40,452,536	96.77	88.32	0.52
KRAS P5	37,886,040	96.53	87.55	0.64
KRAS P5_v2	43,166,706	97.72	88.09	0.32

Overall alignment counts individual reads. Unique and discordant percentages count input pairs; unique means concordantly aligned to one location. Total input: 727,731,715 pairs. The earlier million-pair pilots are excluded.

KRAS P5 is SRR24828471; P5_v2 is SRR24828472. These two MiaPaca-2 runs remain separate while their repeat type is unresolved.

Read quality

All 34 files contain 150-base reads and pass base-quality and adapter checks. Base composition and duplication are flagged in every file. MultiQC reports 57.0%-67.3% duplication. Both SRR24828471 mates also have GC-distribution warnings.

Library preparation can affect base composition, while abundant transcripts and amplification can produce repeated sequences. The quality flags do not identify their cause. No trimming or automatic duplicate removal was applied on the basis of these flags. The GC warning alone does not establish contamination.

Pilot comparison

The six pilots each used the first 1,000,000 read pairs from Control M1 or KRAS P1. Overall read alignment was 96.63% for Control M1 and 96.40% for KRAS P1 under all three strand settings. Control M1 had 86.40% concordant unique pairs in unstranded mode and 86.44% in FR and RF modes; KRAS P1 had 87.43% under each setting.

The similar FR and RF rates did not determine strand specificity; the RSeQC results on page 1 supported the selected setting.

KRAS M1 and next steps

KRAS M1 has 45,292,678 input pairs, but only 1,371 align concordantly to a unique location (0.0030%). Another 75.56% align discordantly. Its 96.51% overall read alignment rate includes reads that align outside the expected pairing relationship. The other 16 runs have 85.19%-88.32% concordant unique pairs.

Check the KRAS M1 input identities, read order and paired-end settings before using it for gene counting. Its cause remains unresolved. The extra MiaPaca-2 KRAS run also needs clarification before selecting inputs for differential expression. Treatment comparisons should account for cell line. Gene counts and differential expression results are not yet available.

References

1. Galaxy FastQC reports, MultiQC 165, all 17 full HISAT2 summaries and RSeQC results, recorded on 22 September 2026. Analysis records: github.com/yichuanlin07/pdac-kas-rnaseq.
2. Course slides: Week 2, Quality Control and Alignment.
3. Klomp JA et al. Defining the KRAS- and ERK-dependent transcriptome in KRAS-mutant cancers. Science. 2024. doi:10.1126/science.adk0775.
4. FastQC documentation. bioinformatics.babraham.ac.uk/projects/fastqc/.
5. HISAT2 manual. daehwankimlab.github.io/hisat2/manual/.
6. RSeQC documentation. rseqc.sourceforge.net.

Author contribution

Yichuan Lin organized the data and coordinated the analysis and report preparation.

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